

Experiment/ Case Study No.: 4

1. **Experiment Title:** Use Function Types, Scope and Closures; Apply Try-Catch; Build a Palindrome Checker

2. **Software/Tools Required:** VS Code, Web Browser, HTML, CSS, JavaScript

3. **Experiment Program Code:**

index.html

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Palindrome Checker</title>
  <link rel="stylesheet" href="style.css">
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;600&display=swap" rel="stylesheet">
</head>

<body>
  <div class="container">
    <h1>Palindrome Checker</h1>
    <p>Enter:</p>
    <div class="input-group">
      <input type="text" id="text-input" placeholder="Enter text here...">
      <button id="check-btn">Check</button>
    </div>
    <div id="result" class="result"></div>
  </div>
  <script src="palindrome.js"></script>
</body>

</html>
```

palindrome.js

```
document.addEventListener('DOMContentLoaded', () => {
  const checkBtn = document.getElementById('check-btn');
  const textInput = document.getElementById('text-input');
  const resultElement = document.getElementById('result');

  checkBtn.addEventListener('click', checkPalindrome);

  textInput.addEventListener('keydown', (e) => {
    if (e.key === 'Enter') {
      checkPalindrome();
    }
  });

  function checkPalindrome() {
    const inputStr = textInput.value;

    resultElement.className = 'result';
    resultElement.innerHTML = '';

    try {
      if (inputStr.trim() === '') {
        throw new Error("Please enter a value to check.");
      }

      const cleanedStr = inputStr.replace(/[^A-Za-z0-9]/g, '').toLowerCase();

      if (cleanedStr === '') {
        throw new Error("Input must contain alphanumeric characters.");
      }

      const reversedStr = cleanedStr.split('').reverse().join('');

      if (cleanedStr === reversedStr) {
        resultElement.innerHTML = `<strong>${inputStr}</strong> is a palindrome.`;
        resultElement.classList.add('success');
      } else {
        resultElement.innerHTML = `<strong>${inputStr}</strong> is not a palindrome.`;
        resultElement.classList.add('error');
      }
    } catch (error) {
      resultElement.innerHTML = `<strong>Invalid Input:</strong> ${error.message}`;
      resultElement.classList.add('invalid');
    }
  }
});
```

style.css

```
body {
  font-family: Arial, sans-serif;
```

```
background-color: #e8f5e9; /* Soft light green matching the theme */
color: #333333;
display: flex;
justify-content: center;
align-items: center;
height: 100vh;
margin: 0;
}

.container {
background-color: #ffffff;
padding: 32px;
width: 400px;
border: 1px solid #c8e6c9;
border-radius: 4px;
}

h1 {
font-size: 24px;
color: #2e7d32; /* Deep solid green */
margin-top: 0;
margin-bottom: 8px;
}

p {
font-size: 14px;
color: #555555;
margin-bottom: 24px;
line-height: 1.5;
}

.input-group {
display: flex;
flex-direction: column;
margin-bottom: 16px;
}

input[type="text"] {
padding: 12px;
font-size: 16px;
border: 1px solid #ced4da;
margin-bottom: 16px;
border-radius: 4px;
}

input[type="text"]:focus {
outline: none;
border-color: #4caf50;
}

button {
padding: 12px;
font-size: 16px;
color: #ffffff;
background-color: #4caf50; /* Standard pleasant green */
border: none;
cursor: pointer;
border-radius: 4px;
font-weight: bold;
}

button:hover {
background-color: #388e3c;
}

.result {
font-size: 16px;
padding: 12px;
margin-top: 16px;
display: none; /* Hidden by default */
border-radius: 4px;
}

/* Showing the result box */
.result.success,
.result.error,
.result.invalid {
display: block;
}

.result.success {
background-color: #d4edda;
color: #155724;
border: 1px solid #c3e6cb;
}
```

```
.result.error {
  background-color: #f8d7da;
  color: #721c24;
  border: 1px solid #f5c6cb;
}

.result.invalid {
  background-color: #fff3cd;
  color: #856404;
  border: 1px solid #ffeeba;
}
```

4. Output:



5. Case Study Title: Vehicle Registration Number Validation Webpage using Functions, Scope, and Try-Catch

6. Case Study Program Code:### vehicle.css

```
body {
  font-family: 'Inter', sans-serif;
  background-color: #e8f5e9; /* Light green background */
  color: #1b5e20;
  display: flex;
  justify-content: center;
  align-items: center;
  height: 100vh;
  margin: 0;
}

.container {
  background-color: #ffffff;
  padding: 2rem;
  border-radius: 8px;
  box-shadow: 0 4px 6px rgba(0,0,0,0.1);
  text-align: center;
  border: 2px solid #4caf50; /* Classic green border */
  width: 350px;
}

h1 {
  color: #2e7d32;
  margin-top: 0;
  margin-bottom: 10px;
  font-size: 1.5rem;
}

p {
  font-size: 0.9rem;
  color: #388e3c;
  margin-bottom: 20px;
}

.input-group {
  display: flex;
  flex-direction: column;
  gap: 10px;
}

input[type="text"] {
  padding: 10px;
  border: 1px solid #81c784;
  border-radius: 4px;
  font-size: 1rem;
  outline: none;
  text-transform: uppercase;
}

input[type="text"]:focus {
  border-color: #388e3c;
  box-shadow: 0 0 5px rgba(76, 175, 80, 0.4);
}

button {
  padding: 10px;
  background-color: #4caf50;
  color: white;
  border: none;
  border-radius: 4px;
  cursor: pointer;
  font-size: 1rem;
  font-weight: 600;
}

button:hover {
  background-color: #45a049;
}
```

```

.hidden {
  display: none;
}

.success {
  margin-top: 20px;
  padding: 10px;
  border-radius: 4px;
  font-weight: 600;
  background-color: #c8e6c9;
  color: #2e7d32;
  border: 1px solid #a5d6a7;
}

.error {
  margin-top: 20px;
  padding: 10px;
  border-radius: 4px;
  font-weight: 600;
  background-color: #ffedd2;
  color: #c62828;
  border: 1px solid #ef9a9a;
}

```

vehicle.html

```

<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Vehicle Registration Number Validation</title>

  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #e8f5e9;
      color: #1b5e20;
      display: flex;
      justify-content: center;
      align-items: center;
      height: 100vh;
      margin: 0;
    }

    .container {
      background-color: #ffffff;
      padding: 30px;
      border-radius: 8px;
      box-shadow: 0 4px 6px rgba(0, 0, 0, 0.15);
      text-align: center;
      border: 2px solid #4caf50;
      width: 380px;
    }

    h1 {
      color: #2e7d32;
      margin-top: 0;
      margin-bottom: 8px;
      font-size: 1.5rem;
    }

    p {
      font-size: 0.9rem;
      color: #388e3c;
      margin-bottom: 18px;
    }

    .format-hint {
      font-size: 0.8rem;
      color: #555;
      margin-bottom: 15px;
    }

    .input-group {
      display: flex;
      flex-direction: column;
      gap: 10px;
    }

    input[type="text"] {
      padding: 10px;
      border: 1px solid #81c784;
      border-radius: 4px;
    }
  </style>

```

```

        font-size: 1rem;
        outline: none;
        text-transform: uppercase;
        text-align: center;
        letter-spacing: 2px;
    }

    input[type="text"]:focus {
        border-color: #388e3c;
        box-shadow: 0 0 5px rgba(76, 175, 80, 0.4);
    }

    button {
        padding: 10px;
        background-color: #4caf50;
        color: white;
        border: none;
        border-radius: 4px;
        cursor: pointer;
        font-size: 1rem;
        font-weight: bold;
    }

    button:hover {
        background-color: #388e3c;
    }

    .result {
        margin-top: 20px;
        padding: 12px;
        border-radius: 4px;
        font-weight: bold;
        display: none;
    }

    .success {
        background-color: #c8e6c9;
        color: #2e7d32;
        border: 1px solid #a5d6a7;
    }

    .error {
        background-color: #ffcdd2;
        color: #c62828;
        border: 1px solid #ef9a9a;
    }

    .rules {
        margin-top: 20px;
        text-align: left;
        background-color: #f1f8f1;
        border: 1px solid #c8e6c9;
        border-radius: 4px;
        padding: 12px 15px;
    }

    .rules h3 {
        margin: 0 0 8px 0;
        color: #2e7d32;
        font-size: 0.9rem;
    }

    .rules ul {
        margin: 0;
        padding-left: 18px;
        font-size: 0.8rem;
        color: #388e3c;
    }

    .rules ul li {
        margin-bottom: 4px;
    }
</style>
</head>

<body>
    <div class="container">
        <h1>Vehicle Registration Validator</h1>
        <p>Enter your Vehicle Registration Number:</p>
        <p class="format-hint">Format: <strong>SS DD AA NNNN</strong> (e.g., MH12AB1234)</p>

        <div class="input-group">
            <input type="text" id="vehicleInput" placeholder="e.g. MH12AB1234" maxlength="10">
            <button id="validate-btn" onclick="validateVehicleReg()">Validate</button>
        </div>
    </div>

```

```

<div id="resultMessage" class="result"></div>

<div class="rules">
  <h3>Validation Rules:</h3>
  <ul>
    <li><strong>Chars 1-2:</strong> Uppercase letters (State Code)</li>
    <li><strong>Chars 3-4:</strong> Digits (District Code)</li>
    <li><strong>Chars 5-6:</strong> Uppercase letters (Series)</li>
    <li><strong>Chars 7-10:</strong> Digits (Vehicle Number)</li>
    <li>Total length must be exactly <strong>10</strong> characters</li>
  </ul>
</div>
</div>

<script>
  function validateVehicleReg() {
    const input = document.getElementById("vehicleInput").value.toUpperCase().trim();
    const resultElement = document.getElementById("resultMessage");

    try {
      if (input === "") {
        throw new Error("Registration number should not be empty.");
      }

      if (input.length !== 10) {
        throw new Error(
          "Length must be exactly 10 characters. You entered " + input.length + " character(s)."
        );
      }

      const stateCode = input.substring(0, 2);
      if (!/^[A-Z]{2}$/.test(stateCode)) {
        throw new Error(
          "First two characters must be uppercase alphabets (State Code). Found: '" + stateCode + "'"
        );
      }

      const districtCode = input.substring(2, 4);
      if (!/^\d{2}$/.test(districtCode)) {
        throw new Error(
          "Characters 3-4 must be digits (District Code). Found: '" + districtCode + "'"
        );
      }

      const series = input.substring(4, 6);
      if (!/^[A-Z]{2}$/.test(series)) {
        throw new Error(
          "Characters 5-6 must be uppercase alphabets (Series). Found: '" + series + "'"
        );
      }

      const vehicleNumber = input.substring(6, 10);
      if (!/^\d{4}$/.test(vehicleNumber)) {
        throw new Error(
          "Last four characters must be digits (Vehicle Number). Found: '" + vehicleNumber + "'"
        );
      }

      resultElement.textContent = "✓ VALID: '" + input + "' is a valid vehicle registration number.";
      resultElement.className = "result success";
      resultElement.style.display = "block";

    } catch (err) {
      resultElement.textContent = "✗ INVALID: " + err.message;
      resultElement.className = "result error";
      resultElement.style.display = "block";
    }
  }

  document.getElementById("vehicleInput").addEventListener("keydown", function (event) {
    if (event.key === "Enter") {
      validateVehicleReg();
    }
  });
</script>
</body>
</html>

```

vehicle.js

```

function validateVehicleReg() {
  const input = document.getElementById("vehicleInput").value;
  const resultElement = document.getElementById("resultMessage");

  try {
    if (!input) {
      throw new Error("Registration number should not be empty.");
    }

    if (input.length !== 10) {
      throw new Error("Length should be exactly 10 characters.");
    }

    const stateCode = input.substring(0, 2);
    if (!/^[A-Z]{2}$/.test(stateCode)) {
      throw new Error("First two characters must be uppercase alphabets (State Code).");
    }

    const districtCode = input.substring(2, 4);
    if (!/^[0-9]{2}$/.test(districtCode)) {
      throw new Error("Next two characters must be digits (District Code).");
    }

    const series = input.substring(4, 6);
    if (!/^[A-Z]{2}$/.test(series)) {
      throw new Error("Next two characters must be uppercase alphabets (Series).");
    }

    const vehicleNumber = input.substring(6, 10);
    if (!/^[0-9]{4}$/.test(vehicleNumber)) {
      throw new Error("Last four characters must be digits (Vehicle Number).");
    }

    resultElement.textContent = "Valid: Registration number is correct.";
    resultElement.className = "success";
  } catch (err) {
    resultElement.textContent = "Invalid: " + err.message;
    resultElement.className = "error";
  }
}

```

7. Output:

8. Result/Conclusion:

The practical was completed successfully. Function types (Function Declaration, Function Expression, Arrow Function), scope (global, function, block), closures, and try-catch exception handling were implemented. A palindrome checker (Task 4.a) and a vehicle registration number validator (Task 4.b) were developed using JavaScript to validate user input and demonstrate string manipulation and error-handling concepts.